# AMATH 732 Lecture: Multiple Scales

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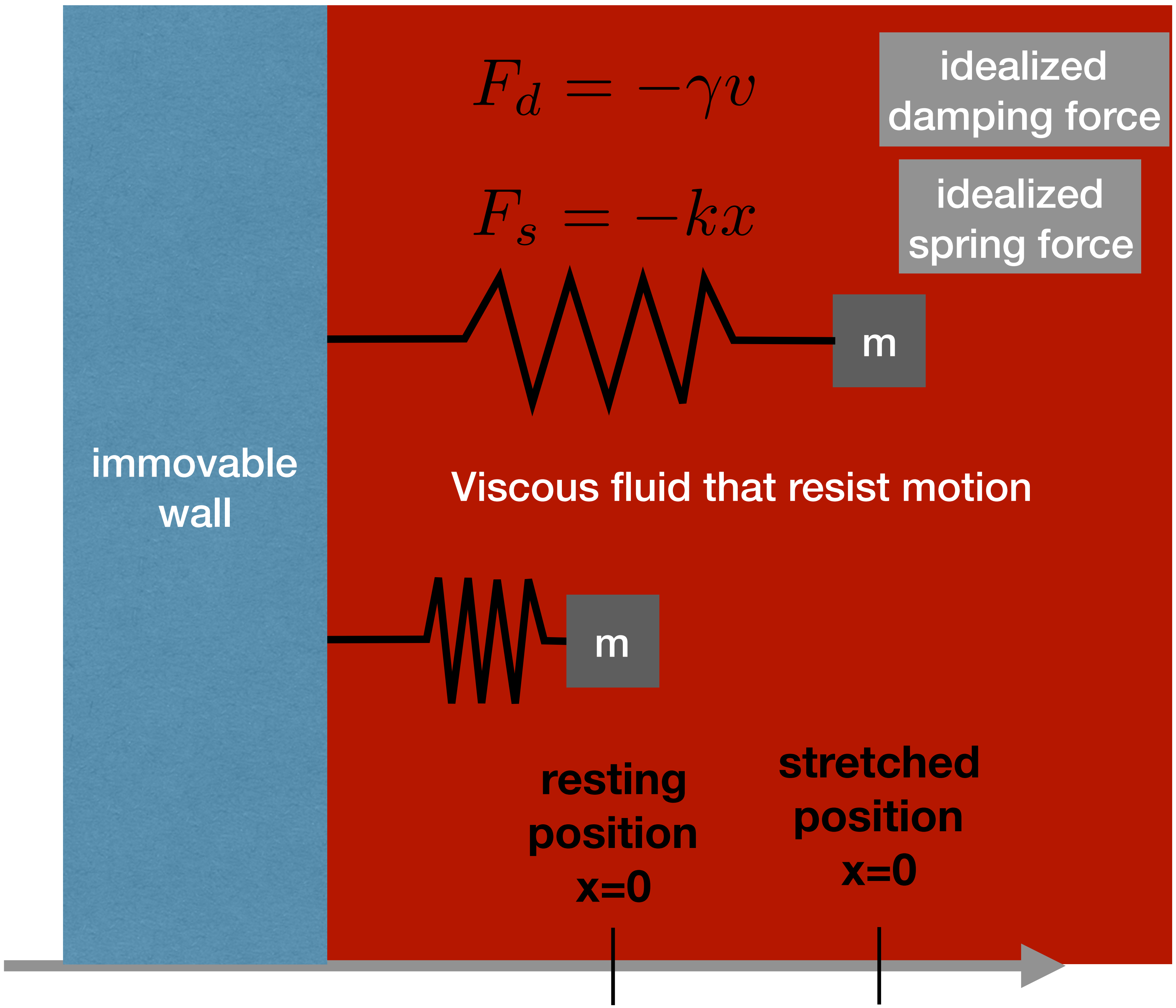
- The purpose of this lecture is three fold.
- First we introduce review the exact solution of the damped SHO and identify multiple time scales.
- Second we demonstrate how RPT fails for this problem.
- Third we introduce the method of multiple scales and give two examples of its use.
- The philosophy of the method is, what intuitively useful results can perturbation theory provide us with before we do numerics.

$$m\frac{d^2x}{dt^2} = -kx - \gamma\frac{dx}{dt}$$

Newton's Second Law



- The damped SHO keeps the assumptions of the SHO but adds:

- The damping is linear and proportional to velocity.

- Energy is defined as before and the first integral derivation follows the same mathematical pattern.

- But energy is no longer conserved.

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = mv\frac{dv}{dt} \qquad \frac{d}{dt}\left(\frac{1}{2}kx^2\right) = kxv$$

$$\frac{d}{dt}\left(E_k + U\right) = -\gamma v^2$$

- Unless the mass is not moving, energy is always being taken out of the system.

- The rate of energy withdrawal is proportional to  the kinetic energy and not the velocity.

- The damped SHO has three types of solutions.
- We focus on the underdamped case:

$$\omega_d = \frac{\sqrt{4mk - \gamma^2}}{2m}$$

$$x(t) = a\exp(-\gamma t/2m)\cos(\omega_d t + \phi)$$

- Again a is the amplitude and phi the phase shift.
- The decay rate depends only on the damping parameter.
- The frequency of oscillation depends on all the physical parameters.

- What about the case in which the damping is weak (assume mass is fixed)?

$$\omega_d = \frac{\sqrt{4mk - \gamma^2}}{2m}$$

$$x(t) = a\exp(-\gamma t/2m)\cos(\omega_d t + \phi)$$

- The damping has no O(1) term and in fact its only non-zero term is at O(gamma).

- The frequency has an O(1) which matches the natural frequency of the undamped SHO.

- The frequency has no O(gamma) correction but there is a correction at O(gamma^2).

- What about the case in which the damping is weak?

$$\omega_d = \frac{\sqrt{4mk - \gamma^2}}{2m}$$

$$x(t) = a\exp(-\gamma t/2)\cos(\omega_d t + \phi)$$

- The damping sets one time scale, 1/gamma, which we know a priori to be long (because gamma is small).

- The frequency sets a second time scale which is set more or less independently of the damping..

- Thus even a problem as simple as the SHO has two inherently different time scales.

- Let's try RPT on the same version of the weakly damped SHO as in the notes.

$$\frac{d^2x}{dt^2} + 2\epsilon\frac{dx}{dt} + x = 0$$

$x(0) = 1$ **and** $x'(0) = v(0) = 0$

$$x(t) = x_0(t) + \epsilon x_1(t) + O(\epsilon^2)$$

- At leading order recover the undamped SHO so get a solution:

$$x_0(t) = \cos(t)$$

- At first order the governing equation reads:

$$\frac{d^2x_1}{dt^2} + x_1 = -2\frac{dx_0}{dt} = 2\sin(t)$$

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- But this equation has no damping either, and so its general solution is:

$$x_1(t) = c_1\cos(t) + c_2\sin(t)$$

- The second term matches the RHS meaning a secular solution. A bit of algebra shows:

$$x_1(t) = \sin(t) - t\cos(t)$$

- So to first order we have:

$$x(t) = \cos(t) + \epsilon\sin(t) - \epsilon t\cos(t)$$

- And for t>1/epsilon the second term is larger than the first.

- This invalidates the assumption of a perturbation expansion and hence **RPT Fails**.

- What went wrong?

- Certainly the algebra is sound so it is not a question of mathematical trickery.

- This means the RPT is actually missing something vital to the problem.

- But we already know what that is!  At each order the ODE RPT gives has only a single natural time scale and the time scale due to damping is absent.

- Put another way, RPT tries to approximate a dissipative system by a series made up of conservative systems.

- This is a recipe for certain failure.

- What to do about it?

- Let's introduce an additional slow timescale: tau=epsilon t

$$\frac{d^2x}{dt^2} + 2\epsilon\frac{dx}{dt} + x = 0$$

$x(0) = 1$ **and** $x'(0) = v(0) = 0$

$$x(t) = x_0(t,\tau) + \epsilon x_1(t,\tau) + O(\epsilon^2)$$

- The trick is that x is now a function of two variables so taking d/dt is really a Chain Rule operation (see below).

- This seems a little bit of a crazy thing to do at first.

- But many mathematical methods start out as a little crazy.

$$\frac{dx}{dt} = \frac{\partial x_0}{\partial t} + \epsilon\left(\frac{\partial x_0}{\partial \tau} + \frac{\partial x_1}{\partial t}\right) + O(\epsilon^2) \qquad \frac{d^2x}{dt^2} = \frac{\partial^2 x_0}{\partial t^2} + \epsilon\left(\frac{\partial^2 x_0}{\partial t \partial \tau} + \frac{\partial^2 x_1}{\partial t^2}\right) + O(\epsilon^2)$$

- At leading order the undamped SHO is recovered, so the general solution is:

$$x_0(t,\tau) = a(\tau)\cos(t) + b(\tau)\sin(t)$$

- Applying the ICs isn't quite so easy now:

$$1 = x(0) = x_0(0{,}0) + \epsilon x_1(0{,}0) + O(\epsilon^2)$$

$$0 = v(0) = \left.\frac{\partial x_0}{\partial t}\right|_{(0,0)} + \epsilon \left.\left(\frac{\partial x_0}{\partial \tau} + \frac{\partial x_1}{\partial t}\right)\right|_{(0,0)} + O(\epsilon^2)$$

- At leading order all we know is a(0)=1 and b(0)=0.

- So we have to go to first order. A bit of algebra shows:

$$\frac{\partial^2 x_1}{\partial t^2} + x_1 = -2\frac{\partial^2 x_0}{\partial \tau \partial t} - 2\frac{\partial x_0}{\partial t}$$

$$\frac{\partial^2 x_1}{\partial t^2} + x_1 = -2\frac{\partial^2 x_0}{\partial \tau \partial t} - 2\frac{\partial x_0}{\partial t}$$

- At first sight this looks as bad as RPT!

- The LHS is again the undamped SHO. But there is a difference that becomes apparent when we substitute what we know about the leading order solution into the RHS.

$$RHS = 2\frac{\partial a}{\partial \tau}\sin(t) - 2\frac{\partial b}{\partial \tau}\cos(t) + 2a(\tau)\sin(t) - 2b(\tau)\cos(t)$$

- If we demand that the coefficients of both sin(t) and cos(t) vanish then we get two ODEs for a(tau) and b(tau).

- This is called **eliminating the secular terms**. Here it gives.

$$\frac{da}{d\tau} = -a \text{ and } \frac{db}{d\tau} = b$$

- Solving and imposing the ICs we get: $a(\tau) = \exp(-\tau); b(\tau) = 0$

- This means the Method of Multiple Scales has given us:

$$x(t) = \exp(-\epsilon t)\cos(t)$$

- This clearly captures the essence of the two scales in the problem.
- It is not quite the same tidy solution RPT gives (when it works).
- Indeed extending to higher order has the potential for very messy algebra.
- But if one eventually solves numerically anyway, it is certainly true that the Method of Multiple Scales gave us intuition for the solution without too much labour.

$$x(t) = \exp(-\epsilon t)\left[\frac{\epsilon}{\sqrt{1-\epsilon^2}}\sin(\sqrt{1-\epsilon^2}t) + \cos(\sqrt{1-\epsilon^2}t)\right]$$

$$x_{series} = \exp(-\epsilon t)\left[\cos(t) + \epsilon\sin(t)\right]$$

- Let’s think a bit about the challenges of the method of multiple scales.

1. As the name suggests, the first is the identification of just how many multiple scales need to be considered. For problems done by hand it is usually just two.

2. The second is what to do about the equations that come from the elimination of secular terms. In the previous example we solved them exactly. This may not always be possible.

- Indeed in some applications the secular equations can supersede the original problem (e.g. wave interaction problems).

- Let's consider a problem we do not have an exact solution for, and one that doesn't dissipate energy.

$$\frac{d^2x}{dt^2} + x + \epsilon x^3 = 0$$

$x(0) = 1$ **and** $x'(0) = v(0) = 0$

$$x(t) = x_0(t,\tau) + \epsilon x_1(t,\tau) + O(\epsilon^2)$$

- The second time scale now comes from the nonlinear term.

- We don't really know ahead of time what it is, nor do we have an exact solution to compare against.

$$\frac{dx}{dt} = \frac{\partial x_0}{\partial t} + \epsilon\left(\frac{\partial x_0}{\partial \tau} + \frac{\partial x_1}{\partial t}\right) + O(\epsilon^2)$$

$$\frac{d^2x}{dt^2} = \frac{\partial^2 x_0}{\partial t^2} + \epsilon\left(\frac{\partial^2 x_0}{\partial t \partial \tau} + \frac{\partial^2 x_1}{\partial t^2}\right) + O(\epsilon^2)$$

- At leading order the undamped SHO is recovered, so the general solution is again:

$$x_0(t,\tau) = a(\tau)\cos(t) + b(\tau)\sin(t)$$

- Applying the ICs isn quite so easy now:

$$1 = x(0) = x_0(0,0) + \epsilon x_1(0,0) + O(\epsilon^2)$$

$$0 = v(0) = \left.\frac{\partial x_0}{\partial t}\right|_{(0,0)} + \epsilon \left.\left(\frac{\partial x_0}{\partial \tau} + \frac{\partial x_1}{\partial t}\right)\right|_{(0,0)} + O(\epsilon^2)$$

- At leading order all we know is a(0)=1 and b(0)=0.

- So we have to go to first order. A bit of algebra shows:

$$\frac{\partial^2 x_1}{\partial t^2} + x_1 = -2\frac{\partial^2 x_0}{\partial t \partial \tau} - x_0^3$$

$$\frac{\partial^2 x_1}{\partial t^2} + x_1 = -2\frac{\partial^2 x_0}{\partial t \partial \tau} - x_0^3$$

- Substitution of the O(1) solution will give several different terms.

- The cubic trig functions can be broken up using standard identities (I used Maple) to get terms like sin(t) and terms like sin(3t) (ditto for cosine).

- Only the sin(t) terms lead to resonance, so that the secular equations read

Am happy to discuss this as part of your own work

$$\frac{da}{d\tau} = \frac{3}{8}b(\tau)\left(a(\tau)^2 + b(\tau)^2\right)$$

$$\frac{db}{d\tau} = -\frac{3}{8}a(\tau)\left(a(\tau)^2 + b(\tau)^2\right)$$

- Notice the common factor which will cancel if we divide:

$$\frac{da}{db} = -\frac{b}{a}$$

$a^2 + b^2 = c_1$ **and from ICs get** $a^2 + b^2 = 1$

$$\left.\begin{array}{c}\dfrac{da}{d\tau} = \dfrac{3}{8}b(\tau) \\ \dfrac{db}{d\tau} = -\dfrac{3}{8}a(\tau) \\ a(0) = 1; b(0) = a'(0) = 0\end{array}\right\} \Rightarrow \quad a(\tau) = \cos(3\tau/8); b(\tau) = -\sin(3\tau/8)$$

- Solving the 1st order DEs gives a(tau) and b(tau) so that the Method of Multiple Scales solution can be recovered:

$$x(t) = \cos\left(\frac{3}{8}\epsilon t\right)\cos(t) - \sin\left(\frac{3}{8}\epsilon t\right)\sin(t) = \cos\left[\left(1 + \frac{3}{8}\epsilon\right)t\right]$$

- A bit of algebra shows that we have a cosine with a modified frequency.

- Thus the effect of the weak nonlinearity is to modify the frequency.

- Changes of shape in phase space would require higher order extensions, or numerical methods.

- Let’s summarize:

1. Many problems have more than one time scale (for BVPS, and PDEs especially, space scales as well).

2. Ignoring the existence of multiple scales leads to the failure of RPT.

3. The Method of Multiple Scales (MMS) fixes what goes wrong with RPT but at the cost of introducing additional independent variables.

4. The MMS effectively reproduces the “essence” of exact solutions, and can provide intuition into solutions when exact solutions are unavailable.

5. The MMS is not easy to turn into an automatic algorithm, but the algebra to first order is generally tractable.

   - Other methods to handle the multiple scales exist, such as the Poincaré-Lindstedt method or the Method of Averaging.

$$\frac{\partial^2 x_1}{\partial t^2} + x_1 = -2\frac{\partial^2 x_0}{\partial t \partial \tau} - x_0^3$$

- Substitution of the O(1) solution gives:

$$RHS = -2\left(-a'(\tau)\cos(t) + b'(\tau)\sin(t)\right) - (a(\tau)\cos(t) + b(\tau)\sin(t))^3$$

$$(a(\tau)\cos(t) + b(\tau)\sin(t))^3 = M_a\cos(t) + M_b\sin(t) + N_a\cos(3t) + N_b\sin(3t)$$

- I can either use trigonometric formulae, like: $\cos(3x) = 4\cos^3(x) - 3\cos(x)$
- Or Maple's expand command, followed by the sneaky "combine(myfunction,trig)" command.
- In either case we only need to find $M_a, M_b$ because these are coefficients of the secular terms that we wish to eliminate.